

Package ‘cramerdb’

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Type Package

Title Simple API → Data Frame (and PostGIS-ready) for Cramer DB

Version 0.1.0

Description One-call helpers to fetch paginated JSON/GeoJSON APIs into data frames or sf tibbles.

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Encoding UTF-8

Depends R (>= 4.1.0)

Imports httr2, jsonlite, tibble, dplyr, purrr, sf

Suggests keyring

Roxygen list(markdown = TRUE)

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browse_endpoints	<i>Browse API endpoints interactively</i>
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Description

Uses interactive filtering (if gum is available) to browse and explore API endpoints. Falls back to listing all endpoints if gum is not available.

Usage

```
browse_endpoints(
  base_url = "https://cramerdbs.com/api/",
  path = NULL,
  staging = FALSE
)
```

Arguments

base_url	Character. Base API URL (default: "https://cramerdbs.com/api/")
path	Character. Optional starting path (e.g., "seine")
staging	Logical. If TRUE, routes requests to the staging server.

Value

Selected endpoint URL or NULL if cancelled

Examples

```
## Not run:
# Interactive browsing (requires gum)
browse_endpoints()

# Start from a specific path
browse_endpoints(path = "seine")

## End(Not run)
```

clear_token	<i>Clear stored CramerDB API token</i>
-------------	--

Description

Removes the token from both the current R session and the system keyring.

Usage

```
clear_token()
```

Value

Invisibly returns TRUE

Examples

```
## Not run:
clear_token()

## End(Not run)
```

create	Create new records via API (POST)
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Description

Create new records via API (POST)

Usage

```
create(
  url,
  data,
  headers = list(),
  id_col = "id",
  style = c("auto", "plain", "feature"),
  chunk_size = 200L,
  base_url = "https://cramerdbs.com/api/",
  dry_run = FALSE,
  staging = FALSE
)
```

Arguments

url	Endpoint - full URL or relative path (e.g., "seine/event/" or "https://cramerdbs.com/api/seine/event/")
data	data.frame or sf tibble
headers	named list of HTTP headers (e.g., list(Authorization = "Token ..."))
id_col	primary key col (default "id"). If missing or NA, server may assign.
style	"auto", "plain", or "feature" (GeoJSON Feature). "auto" picks "feature" for sf, else "plain".
chunk_size	number of rows per request batch (each row is one request; chunking controls progress/messages)
base_url	Character. Base API URL (default: "https://cramerdbs.com/api/")
dry_run	Logical. If TRUE, show what would be sent without actually sending. Useful for validation.
staging	Logical. If TRUE, routes requests to the staging server.

endpoints*List available API endpoints*

Description

Navigate the CramerDB API tree structure by listing available endpoints. Use `whoami()` to check authentication status.

Usage

```
endpoints(  
  path = NULL,  
  base_url = "https://cramerdbs.com/api/",  
  headers = list(),  
  staging = FALSE  
)
```

Arguments

<code>path</code>	Character. Optional API path segment (e.g., "seine", "biology"). If NULL or missing, lists the root API endpoints.
<code>base_url</code>	Character. Base API URL (default: "https://cramerdbs.com/api/")
<code>headers</code>	Named list of HTTP headers (e.g., <code>list(Authorization = "Token xxx")</code>)
<code>staging</code>	Logical. If TRUE, routes requests to the staging server.

Value

Invisibly returns the parsed JSON response; prints formatted output

Examples

```
## Not run:  
# List top-level endpoints  
endpoints()  
  
# List endpoints for a specific section  
endpoints("seine")  
endpoints("biology")  
  
# Equivalent to:  
endpoints("biology/some-sub-path")  
  
## End(Not run)
```

fetch	<i>Fetch any Cramer API endpoint into a tibble</i>
-------	--

Description

fetch() follows pagination automatically and normalizes both plain JSON lists and GeoJSON FeatureCollections into a plain tibble.

Usage

```
fetch(  
  url,  
  headers = list(),  
  base_url = "https://cramerdb.com/api/",  
  staging = FALSE,  
  query = list()  
)
```

Arguments

url	Character. Full URL or relative path (e.g., "tagging/event/"). Relative paths are prepended with base_url.
headers	Named list of HTTP headers (e.g., list(Authorization = "Token xxx")).
base_url	Character. Base API URL (default: "https://cramerdb.com/api/")
staging	Logical. If TRUE, routes requests to the staging server.
query	Named list of query parameters appended to the URL.

Value

A tibble.

fields	<i>List field names at an API endpoint</i>
--------	--

Description

Returns a character vector of field names available at the given endpoint. First attempts an OPTIONS request to read the DRF schema; falls back to fetching one record and reading its keys.

Usage

```
fields(  
  path,  
  base_url = "https://cramerdb.com/api/",  
  headers = list(),  
  staging = FALSE  
)
```

Arguments

path	Character. Endpoint path (e.g., "seine/event/").
base_url	Character. Base API URL (default: "https://cramerdbs.com/api/")
headers	Named list of HTTP headers.
staging	Logical. If TRUE, routes to the staging server.

Value

A character vector of field names (visible), or NULL on failure.

get_token	<i>Get the currently configured CramerDB API token</i>
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Description

Retrieves the token from (in order of priority):

1. R options (current session)
2. System keyring (if available)
3. CRAMERDB_TOKEN environment variable

Usage

```
get_token(error_if_missing = FALSE)
```

Arguments

error_if_missing	Logical. If TRUE, error when no token is found.
------------------	---

Value

The token string, or NULL if not set (and error_if_missing = FALSE).

Examples

```
## Not run:  
get_token()  
  
## End(Not run)
```

`set_token`*Set the CramerDB API token for this R session*

Description

Stores the token securely in your system's credential store (Keychain on macOS, Credential Manager on Windows, Secret Service on Linux). The token persists across R sessions and is retrieved automatically when needed.

Usage

```
set_token(token, persist = TRUE)
```

Arguments

<code>token</code>	Character scalar token value (no "Token " prefix needed).
<code>persist</code>	Logical. If TRUE (default), store in system keyring for persistence across sessions. If FALSE, store only in R options for current session.

Details

If the keyring package is not available, falls back to storing in R options for the current session only.

Examples

```
## Not run:  
# Store token securely (persists across sessions)  
set_token("abcd1234...")  
  
# Session-only storage  
set_token("abcd1234...", persist = FALSE)  
  
## End(Not run)
```

`test_connection`*Test connection to CramerDB API*

Description

Verifies that the API is reachable and your authentication is working. Provides a quick health check before running data operations.

Usage

```
test_connection(base_url = "https://cramerdb.com/api/", staging = FALSE)
```

Arguments

<code>base_url</code>	Character. Base API URL (default: "https://cramerdb.com/api/")
<code>staging</code>	Logical. If TRUE, routes requests to the staging server.

Value

Invisibly returns TRUE if connection successful, FALSE otherwise

Examples

```
## Not run:
test_connection()

## End(Not run)
```

update	<i>Update records via API (PATCH by id)</i>
--------	---

Description

Update records via API (PATCH by id)

Usage

```
update(
  url,
  data,
  headers = list(),
  id_col = "id",
  style = c("auto", "plain", "feature"),
  chunk_size = 200L,
  base_url = "https://cramerdb.com/api/",
  dry_run = FALSE,
  staging = FALSE
)
```

Arguments

url	Endpoint - full URL or relative path (e.g., "seine/event/" or "https://cramerdb.com/api/seine/event/")
data	data.frame or sf tibble
headers	named list of HTTP headers
id_col	primary key col (default "id")
style	"auto", "plain", or "feature"
chunk_size	number of rows per request batch
base_url	Character. Base API URL (default: "https://cramerdb.com/api/")
dry_run	Logical. If TRUE, show what would be sent without actually sending. Useful for validation.
staging	Logical. If TRUE, routes requests to the staging server.

upsert	<i>Upsert records via API (PATCH if id present & exists; otherwise POST)</i>
--------	--

Description

Upsert records via API (PATCH if id present & exists; otherwise POST)

Usage

```
upsert(
  url,
  data,
  headers = list(),
  id_col = "id",
  style = c("auto", "plain", "feature"),
  chunk_size = 200L,
  base_url = "https://cramerdb.com/api/",
  dry_run = FALSE,
  staging = FALSE
)
```

Arguments

url	Endpoint - full URL or relative path (e.g., "seine/event/" or "https://cramerdb.com/api/seine/event/")
data	data.frame or sf tibble
headers	named list of HTTP headers
id_col	primary key col (default "id")
style	"auto", "plain", or "feature"
chunk_size	number of rows per request batch
base_url	Character. Base API URL (default: "https://cramerdb.com/api/")
dry_run	Logical. If TRUE, show what would be sent without actually sending. Useful for validation.
staging	Logical. If TRUE, routes requests to the staging server.

whoami	<i>Check authentication status</i>
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Description

Check if you are authenticated with the CramerDB API and see which user you are logged in as.

Usage

```
whoami(
  base_url = "https://cramerdb.com/api/",
  headers = list(),
  staging = FALSE
)
```

Arguments

<code>base_url</code>	Character. Base API URL (default: "https://cramerdbs.com/api/")
<code>headers</code>	Named list of HTTP headers (e.g., <code>list(Authorization = "Token xxx")</code>)
<code>staging</code>	Logical. If TRUE, routes requests to the staging server.

Value

Invisibly returns the parsed JSON response; prints formatted output

Examples

```
## Not run:  
# Check authentication status  
whoami()  
  
# After setting token  
set_token("your_token_here")  
whoami()  
  
## End(Not run)
```

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